

ORTHOGONAL APPROXIMATION OF EXTREMAL FIELD MAPS

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Orthogonal polynomial approximations are used to build extremal field maps that interpolate solutions to optimal control problems in multi-variable systems. Optimal control problems are solved via an indirect integral collocation method using Chebyshev polynomials, yielding improved accuracy and significant runtime speedups. Solutions in the multi-dimensional parameter space are sampled at nodes which yield discrete orthogonality conditions. Extremal field maps are found in a computationally efficient manner as the trivial to invert least-squares solution which approximates all states, costates, and control variables, and the error between node points is quantified. Bang-bang low-thrust trajectories for elliptical orbit transfers are computed.

INTRODUCTION

In optimal control problems (OCPs), the objective often entails minimizing or maximizing a performance metric subject to some set of constraints imposed by the dynamical system.¹ Solution methods for OCPs are described as being direct or indirect.² While direct methods seek to find the control and state trajectory that minimize the functional through nonlinear programming by discretizing the cost functional, indirect methods derive the governing state and costate equations through Pontryagin's minimum principle, and the control is usually found to be a function of the costates.³ Indirect methods are often preferable for dynamical systems, when feasible to compute, because they yield a continuous solution that is well-described by and rigorously satisfy the derived necessary conditions. The solution yielded by indirect methods can be characterized by a set of initial costates which, when used as the initial conditions for integration of the state and costate equations, yield the full optimal control and trajectory. Due to the shooting methods required to solve the two-point boundary value problem (TPBVP) created by indirect methods, they require significant computation time to yield a solution; therefore, any direct relationship between the parameters of an OCP and the initial costates has great utility.

This mathematical relationship between the parameters and the initial costates is referred to as an *extremal field map*. While an extremal field map cannot be computed analytically, individual solutions to OCPs can be combined to create a discrete map, which, when properly constructed, allows for precise interpolation. These maps can be used in a variety of applications such as autonomous guidance for spacecraft.⁴

For both indirect methods for solving OCPs and the creation of extremal field maps, efficient approximation methods can be used to greatly improve the solution speed. Additionally, for ex-

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tremal field maps, it can allow for a smooth, continuous approximation of the relationship between the parameters and costates. In recent years, many techniques for solving initial value problems (IVPs) and boundary value problems (BVPs) have been developed which use orthogonal function approximation. Due to their spectral convergence properties, an excellent choice of basis functions is Chebyshev polynomials.⁵ Two such methods which use Chebyshev polynomials to solve IVPs and BVPs are Modified Chebyshev-Picard Iteration (MCPI),^{6–8} which has been used to generate extremal field maps by Woollands et. al.,⁹ and Integral Chebyshev Collocation (ICC),¹⁰ the latter of which is used in this work. This solution method is independent from the N -dimensional Chebyshev approximation that is used to interpolate the extremal field maps.

In this work, bang-bang low-thrust trajectory solutions are found using indirect ICC methods. Orthogonal approximation are then used to build the extremal field maps describing a range of trajectories for many possible boundary conditions and system parameter values, over a large domain of interest. Single and multi-dimensional approximations of extremal field maps are created to describe the relationships between parameters and the initial costates which characterize the solution to the OCP.

INTEGRAL CHEBYSHEV COLLOCATION

This section provides a brief introduction to ICC, a novel method for solving IVPs and BVPs through the use of orthogonal function approximation. A more detailed description of ICC and examples IVP and BVP problems are given by Peck.¹⁰

Chebyshev polynomials are a particularly useful family of orthogonal polynomials due to their inherent properties that enhance both near-uniform accuracy and efficiency in numerical analysis. These polynomials are defined over the domain $\tau \in [-1, 1]$ as

$$T_n(\tau) = \cos(n \arccos \tau) \quad (1)$$

and have the term recursion

$$T_n(\tau) = 2\tau T_{n-1}(\tau) - T_{n-2}(\tau), \quad n \geq 2 \quad (2)$$

$$T_0(\tau) = 1 \quad (3)$$

$$T_1(\tau) = \tau \quad (4)$$

An n^{th} order approximation of a scalar function at the point τ , denoted as ${}^n f(\tau)$, is constructed as a linear combination of Chebyshev polynomials

$$f(\tau) \approx {}^n f(\tau) \equiv \sum_{i=0}^n \alpha_i T_i(\tau) = \alpha_0 T_0(\tau) + \alpha_1 T_1(\tau) + \dots + \alpha_n T_n(\tau) \quad (5)$$

$$(6)$$

which are written compactly in vector-matrix form

$${}^n f(\tau) = \begin{bmatrix} T_0(\tau) & T_1(\tau) & \dots & T_n(\tau) \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \quad (7)$$

$${}^n f(\tau) = {}^n \mathbf{T}(\tau) \boldsymbol{\alpha}, \quad {}^n \mathbf{T}(\tau) \in \mathbb{R}^{1 \times n+1}, \quad \boldsymbol{\alpha} \in \mathbb{R}^{n+1 \times 1} \quad (8)$$

where $\boldsymbol{\alpha}$ is a column vector of polynomial coefficients and ${}^n \mathbf{T}(\tau)$ is a row vector of Chebyshev polynomials evaluated at τ , with the pre-superscript denoting the order of the final polynomial in the series.

Defining the set of interpolation nodes $\boldsymbol{\tau} = [\tau_0, \tau_1, \dots, \tau_n]^\top$, the function approximation is extended to all nodes as

$$\begin{bmatrix} f(\tau_0) \\ f(\tau_1) \\ \vdots \\ f(\tau_n) \end{bmatrix} = \begin{bmatrix} T_0(\tau_0) & T_1(\tau_0) & \cdots & T_n(\tau_0) \\ T_0(\tau_1) & T_1(\tau_1) & \cdots & T_n(\tau_1) \\ \vdots & \vdots & \ddots & \vdots \\ T_0(\tau_n) & T_1(\tau_n) & \cdots & T_n(\tau_n) \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \quad (9)$$

$$\mathbf{f}(\boldsymbol{\tau}) = {}^n \mathcal{T}(\boldsymbol{\tau}) \boldsymbol{\alpha}, \quad {}^n \mathcal{T}(\boldsymbol{\tau}) \in \mathbb{R}^{n+1 \times n+1} \quad (10)$$

The nodes $\boldsymbol{\tau}$ can be chosen such that the columns of ${}^n \mathcal{T}(\boldsymbol{\tau})$ are discretely orthogonal. The nodes that achieve this are the roots of the $(n+1)^{\text{th}}$ order polynomial, given in Eq. (11), and are known as the Chebyshev-Gauss (CG) nodes. The CG nodes have been the subject of much recent work and have been shown to be the optimal interpolation nodes.^{11,12} This discrete orthogonality condition leads to the columns of ${}^n \mathcal{T}(\boldsymbol{\tau})$ acting as orthogonal basis vectors.

$$\tau_k = \cos \left(\frac{(k - \frac{1}{2})\pi}{n+1} \right), \quad k = [1, 2, \dots, n+1] \quad (11)$$

To form a solution using ICC, the highest order derivative (i.e., the dynamics) are approximated with a Chebyshev series as in Eq. (10) and is converted from the time domain $t \in [t_0, t_f]$ to the transformed computational domain $\tau \in [-1, 1]$ via

$$\tau = \frac{2}{\Delta t} t - \frac{t_f + t_0}{\Delta t}, \quad \Delta t = t_f - t_0 \quad (12)$$

Integral properties of the Chebyshev polynomials are then leveraged to integrated until reaching the desired state. The first, second, and i^{th} integral of Chebyshev polynomials are given in terms of the adjacent polynomials as^{5,13}

$$\int T_0(\tau) d\tau = T_1(\tau) \quad (13)$$

$$\int T_1(\tau) d\tau = \frac{1}{4} (T_2(\tau) + T_0(\tau)) \quad (14)$$

$$\int T_i(\tau) d\tau = \frac{1}{2(i+1)} T_{i+1}(\tau) + \frac{1}{2(i-1)} T_{i-1}(\tau), \quad i \geq 2 \quad (15)$$

Thus, integrating over the Chebyshev domain from -1 to the nodes $\boldsymbol{\tau}$ gives

$$\int_{-1}^{\boldsymbol{\tau}} {}^n \mathcal{T}(\tau) d\tau = [{}^{n+1} \mathcal{T}(\tau) \mathcal{F}]_{-1}^{\boldsymbol{\tau}} \quad (16)$$

$$= [{}^{n+1} \mathcal{T}(\boldsymbol{\tau}) - {}^{n+1} \mathcal{T}(-1)] \mathcal{F} \quad (17)$$

$$\mathcal{F} = \begin{bmatrix} 0 & \frac{1}{4} & 0 & \cdots & 0 & 0 & 0 \\ 1 & 0 & -\frac{1}{2} & \cdots & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{2(n-1)} & 0 & -\frac{1}{2(n-1)} \\ 0 & 0 & 0 & \cdots & 0 & \frac{1}{2n} & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & \frac{1}{2(n+1)} \end{bmatrix} \in \mathbb{R}^{n+2 \times n+1} \quad (18)$$

where $\mathcal{F}$ is the integration weight matrix composed of entries only on the upper and lower diagonals. The matrix $\mathcal{F}$ can be viewed as an integration operator which projects the integral ${}^n\mathcal{T}$ back onto the Chebyshev basis functions. Note that the order of the series increases by one since the integral of the n^{th} polynomial yields an $(n+1)^{\text{th}}$ order polynomial. However, since the $(n+1)^{\text{th}}$ polynomial is evaluated at its own roots for the upper integration bound, the additional column is all zeros. This can then be rewritten as

$$\int_{-1}^{\tau} {}^n\mathcal{T}(\tau) d\tau = {}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - \mathbf{v}(-1) \quad (19)$$

$$\mathbf{v}(-1) = {}^{n+1}\mathcal{T}(-1)\mathcal{F} \in \mathbb{R}^{1 \times n+1} \quad (20)$$

$$\tilde{\mathcal{F}} = \mathcal{F}_{1:n+1,1:n+1} \in \mathbb{R}^{n+1 \times n+1} \quad (21)$$

where $\mathbf{v}(-1)$ is a constant vector of the integrated polynomials evaluated at the lower bound -1 , and $\tilde{\mathcal{F}}$ is equivalent to $\mathcal{F}$ with the final row removed. This integration scheme can be continued to yield the m^{th} integral¹⁰

$${}^{(m)}\int_{-1}^{\tau} {}^n\mathcal{T}(\tau) d\tau = {}^n\mathcal{T}(\tau)\tilde{\mathcal{F}}^m - \mathbf{v}(-1) \sum_{j=0}^{m-1} \tilde{\mathcal{F}}^{m-1-j} p_j(\tau) \quad (22)$$

$$p_j(\tau) = \frac{1}{j!} \sum_{l=0}^j \binom{j}{l} \tau^{j-l} \quad (23)$$

For IVPs, a Chebyshev series is used for each degree of freedom and the solution is found by finding the values of the unknown coefficients α . Linear differential equations can be solved with a single matrix inversion, while nonlinear problems require an iterative method such as Newton's method. BVPs can also be solved with ICC by approximating the costates with their own Chebyshev series. These solutions are efficient and accurate since the matrices and vectors involved remain unchanged after the approximation order n is selected. The specific procedure for the example problem is outlined in the corresponding section.

GENERATION AND APPROXIMATION OF EXTREMAL FIELD MAPS

For an indirect method, the optimal solution to an OCP is characterized by a set of initial costates that, when used as the initial conditions for integration, provides the complete optimal state and control history. Once a solution to an initial optimal control problem is found, it is useful to also understand the optimal solutions to problems with neighboring initial conditions. This continuous interpolated relation between the parameter space in an optimal control problem and the corresponding optimal solution is an extremal field map. In general, this is a mapping characterized as

$f : \mathbb{R}^j \rightarrow \mathbb{R}^k$, where j is the number of functional dependent parameters being considered, and k is the number of costates.

Because indirect methods for solving optimal control problems require a non-linear solution procedure, an analytical relationship between a given parameter and the costates which describe the optimal solution cannot be found. Therefore, an extremal field map must be created by finding optimal solutions at discrete points in the parameter space being considered and interpolating them smoothly to within suitable precision. These discrete solutions can then be used to approximate a smooth extremal field map over some set of parameters.

In this work, orthogonal approximation is the chosen method used to create smooth approximations of extremal field maps. In particular, the map is approximated as a linear combination of basis functions,

$$f(x) \approx \sum_{i=0}^N a_i \phi_i(x) \quad (24)$$

where $\phi_i(x)$ is a member of a set of linearly independent basis functions $\{\phi_0(x), \phi_1(x), \dots, \phi_N(x)\}$. For a set of points $\mathbf{x} = \{x_0, x_1, \dots, x_M\}$, this is given in matrix notation as

$$\mathbf{f} \approx \Phi(\mathbf{x})\mathbf{a} \quad (25)$$

$$\Phi(\mathbf{x}) = \begin{bmatrix} \phi_0(x_0) & \phi_1(x_0) & \dots & \phi_N(x_0) \\ \phi_0(x_1) & \phi_1(x_1) & \dots & \phi_N(x_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(x_M) & \phi_1(x_M) & \dots & \phi_N(x_M) \end{bmatrix} \quad (26)$$

$$\mathbf{a} = [a_0 \ a_1 \ \dots \ a_N]^\top \quad (27)$$

where $\mathbf{a}$ is the vector of coefficients which minimize the weighted least square error

$$J = \frac{1}{2}(\mathbf{f} - \Phi\mathbf{a})^\top W(\mathbf{f} - \Phi\mathbf{a}) \quad (28)$$

and is determined by the normal equations as¹⁴

$$\mathbf{a} = (\Phi^\top W \Phi)^{-1} \Phi^\top W \mathbf{f} \quad (29)$$

such that W is a $(M+1) \times (M+1)$ positive definite weight matrix. If the basis functions are chosen to be a set of orthogonal basis functions, and W is chosen to be diagonal, then it is possible to choose W and sample node location such that $\Phi^\top W \Phi$ becomes diagonal, and thus trivial to invert. In particular, when the basis functions are chosen to be the Chebyshev polynomials and f is sampled at the polynomial extrema, known as the Chebyshev-Gauss-Lobatto (CGL) nodes, defined as

$$\xi_i = -\left(\frac{i\pi}{M}\right), \quad i = 0, 1, \dots, M \quad (30)$$

then the weight matrix can be chosen to be $W = \text{diag}\{1/2, 1, 1, \dots, 1, 1, 1/2\}$ such that $\Phi^\top W \Phi$ is diagonal. Once the vector of coefficients is determined, the interpolation is given as

$$f(x) \approx \begin{bmatrix} \phi_0(x) & \phi_1(x) & \dots & \phi_N(x) \end{bmatrix} \mathbf{a}, \quad x \in [-1, 1] \quad (31)$$

Additionally, this method expands well into multivariable functions.¹⁵ Analogous to Eq. (24), the formulation for approximating a function of multiple variables can be written as

$$f(\xi, \eta) \approx \sum_{i_x=0}^{N_x} \sum_{i_y=0}^{N_y} a_{i_x i_y} \phi_{i_x i_y}(\xi, \eta) \quad (32)$$

The matrix form $\mathbf{f} = \Phi \mathbf{a}$ is given as

$$\mathbf{f} = \begin{bmatrix} f(\xi_0, \eta_0) \\ \vdots \\ f(\xi_0, \eta_{N_y}) \\ f(\xi_1, \eta_0) \\ \vdots \\ f(\xi_1, \eta_{N_y}) \\ \vdots \\ f(\xi_{N_x}, \eta_0) \\ \vdots \\ f(\xi_{N_x}, \eta_{N_y}) \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_{00} \\ \vdots \\ a_{0N_y} \\ a_{10} \\ \vdots \\ a_{1N_y} \\ \vdots \\ a_{N_x 0} \\ \vdots \\ a_{N_x N_y} \end{bmatrix} \quad (33)$$

and Φ can be expressed simply via the Kronecker product¹⁴

$$\Phi = \Phi_x \otimes \Phi_y \quad (34)$$

where

$$\Phi_x = \begin{bmatrix} \phi_0(\xi_0) & \phi_1(\xi_0) & \cdots & \phi_N(\xi_0) \\ \phi_0(\xi_1) & \phi_1(\xi_1) & \cdots & \phi_N(\xi_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\xi_M) & \phi_1(\xi_M) & \cdots & \phi_N(\xi_M) \end{bmatrix}, \quad \Phi_y = \begin{bmatrix} \phi_0(\eta_0) & \phi_1(\eta_0) & \cdots & \phi_N(\eta_0) \\ \phi_0(\eta_1) & \phi_1(\eta_1) & \cdots & \phi_N(\eta_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\eta_M) & \phi_1(\eta_M) & \cdots & \phi_N(\eta_M) \end{bmatrix} \quad (35)$$

Then the vector of constants can again be found using the normal equations of Eq. (29).

This procedure provides an efficient and powerful method for creating a smooth approximation

$$f(x, y, \dots) \approx \left(\begin{bmatrix} \phi_0(x) & \phi_1(x) & \cdots & \phi_N(x) \end{bmatrix} \otimes \begin{bmatrix} \phi_0(y) & \phi_1(y) & \cdots & \phi_N(y) \end{bmatrix} \otimes \dots \right) \mathbf{a} \quad (36)$$

which can be applied directly to approximate extremal field maps of multiple parameters. Additionally, several approximations can be concatenated to create a map from the full chosen parameter space to the full vector of initial costates.

PLANAR ELLIPTICAL ORBIT TRANSFER

To demonstrate the utility of using orthogonal approximation to generate extremal field maps, a coplanar elliptical orbit transfer problem from Peck¹⁰ is considered. The dynamics of the problem are derived in polar form as the five ordinary differential equations

$$\dot{r} = u \quad (37a)$$

$$\dot{\theta} = \frac{v}{r} \quad (37b)$$

$$\dot{u} = \frac{v^2}{r} - \frac{\mu}{r^2} + \frac{T}{m} \delta \hat{u}_r \quad (37c)$$

$$\dot{v} = \frac{uv}{r} + \frac{T}{m} \delta \hat{u}_\theta \quad (37d)$$

$$\dot{m} = -\frac{T}{c} \delta \quad (37e)$$

where r is the radial position from the central body, θ is the true anomaly polar angle, $\dot{m}$ is the propellant expenditure rate, μ is the gravitational parameter, $c = I_{sp}g_0$ is the exhaust exit velocity, T is the thrust, and $\delta \in [0, 1]$ is the throttle controlled in the $\hat{\mathbf{u}} = [\hat{u}_r, \hat{u}_\theta]^\top$ direction. The cost function for the fixed final time, minimum fuel problem is

$$\min J = \int_{t_0}^{t_f} \frac{T}{c} \delta \, dt \quad (38)$$

$$\text{s.t.} \quad \dot{\mathbf{x}}(t) = \mathbf{f}(t) = \begin{bmatrix} \dot{r} & \dot{\theta} & \dot{u} & \dot{v} & \dot{m} \end{bmatrix}^\top \quad (39)$$

$$\psi(t_f, \mathbf{x}(t_f)) = \mathbf{0} \quad (40)$$

$$\hat{\mathbf{u}}^\top \hat{\mathbf{u}} = 1 \quad (41)$$

where $\psi(t_f, \mathbf{x}(t_f))$ are terminal boundary conditions. These are included to form the augmented cost function and Hamiltonian as

$$\min J_a = \boldsymbol{\nu}^\top \psi(t_f, \mathbf{x}(t_f)) + \int_{t_0}^{t_f} \left(\frac{T}{c} \delta + \boldsymbol{\lambda}^\top (\mathbf{f} - \dot{\mathbf{x}}) \right) dt \quad (42)$$

$$\mathcal{H} = \frac{T}{c} \delta + \boldsymbol{\lambda}^\top \mathbf{f} \quad (43)$$

where $\boldsymbol{\nu}$ are Lagrange multipliers and $\boldsymbol{\lambda}$ are the costates. The necessary conditions yield the costate equations

$$\dot{\boldsymbol{\lambda}} = - \left[\frac{\partial \mathcal{H}}{\partial \mathbf{x}} \right]^\top \quad (44a)$$

$$\dot{\lambda}_r = \lambda_\theta \frac{v}{r^2} + \lambda_u \left(\frac{v^2}{r^2} - \frac{2\mu}{r^3} \right) - \lambda_v \frac{uv}{r^2} \quad (44b)$$

$$\dot{\lambda}_\theta = 0 \quad (44c)$$

$$\dot{\lambda}_u = -\lambda_r + \lambda_v \frac{v}{r} \quad (44d)$$

$$\dot{\lambda}_v = -\lambda_\theta \frac{1}{r} - 2\lambda_u \frac{v}{r} + \lambda_v \frac{u}{r} \quad (44e)$$

$$\dot{\lambda}_m = \frac{T}{m^2} \delta (\lambda_u \hat{u}_r + \lambda_v \hat{u}_\theta) \quad (44f)$$

The transversality conditions describing the final costates are

$$\lambda_r(t_f) = \nu_r \quad (45a)$$

$$\lambda_\theta(t_f) = \nu_\theta \quad (45b)$$

$$\lambda_u(t_f) = \nu_u \quad (45c)$$

$$\lambda_v(t_f) = \nu_v \quad (45d)$$

$$\lambda_m(t_f) = 0 \quad (45e)$$

An additional condition is needed for the (free) final true anomaly f_f

$$-\nu_r \frac{a_f(1 - e_f^2)e_f \sin f_f}{(1 + e_f \cos f_f)^2} - \nu_\theta - \nu_u \frac{\mu e_f \cos f_f}{h_f} + \nu_v \frac{\mu e_f \sin f_f}{h_f} = 0 \quad (46)$$

where a_f is semi-major axis, e_f is eccentricity, and h_f is angular momentum of the final targeted orbit.

In order to use bang-bang control, the switching function S is introduced to the Hamiltonian. Collecting on the terms with throttle gives the function

$$S = \frac{1}{c} (\lambda_m - 1) - \frac{1}{m} (\lambda_u + \lambda_v) \quad (47)$$

Following Pontryagin's principle, the optimal control direction and throttle that minimize the Hamiltonian are

$$\hat{\mathbf{u}} = \frac{\mathbf{p}}{\|\mathbf{p}\|} \quad (48)$$

where $\mathbf{p} = [\lambda_u, \lambda_v]^\top$ is the primer vector.¹⁶ The state and co-state differential equations (37) and (44) constitute a tenth order system. Minimization of the Hamiltonian results in the optimal throttle

$$\delta = \begin{cases} 1, & \text{if } S > 0 \\ 0, & \text{if } S < 0 \end{cases} = \frac{1}{2} [1 + \text{sign}(S)] \quad (49)$$

To avoid needing to detect multiple unknown instantaneous on-off switches of the throttle, hyperbolic tangent smoothing via the parameter ρ is used to convert the discontinuous control to a continuous one^{17,18}

$$\delta(S) = \frac{1}{2} \left[1 + \tanh \left(\frac{S}{\rho} \right) \right], \quad \rho \in [0, 1] \quad (50)$$

Introducing the smooth switching of Eq. (50) and beginning with $\rho = 1$, a sequence of mathematically smooth OCPs that approach the optimal solution with arbitrary precision. Sweeping $\rho \rightarrow 0$ automatically locates the optimal switching times for δ which would result from Eq. (49).

To solve this optimal control problem using ICC, each of the state and costate dynamics are approximated with a Chebyshev series. Since $\lambda_\theta = 0$, λ_θ is a constant determined by the Lagrange multiplier ν_θ . The dynamic and integrated state approximations for the remaining nine states and costates are given in the Chebyshev domain as

$$r'(\tau) = {}^n\mathcal{T}(\tau)\alpha_r, \quad r(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - \mathbf{v}(-1) \right] \alpha_r \quad (51a)$$

$$\theta'(\tau) = {}^n\mathcal{T}(\tau)\alpha_\theta, \quad \theta(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - \mathbf{v}(-1) \right] \alpha_\theta \quad (51b)$$

$$u'(\tau) = {}^n\mathcal{T}(\tau)\alpha_u, \quad u(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - \mathbf{v}(-1) \right] \alpha_u \quad (51c)$$

$$v'(\tau) = {}^n\mathcal{T}(\tau)\alpha_v, \quad v(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - \mathbf{v}(-1) \right] \alpha_v \quad (51d)$$

$$m'(\tau) = {}^n\mathcal{T}(\tau)\alpha_m, \quad m(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - v(-1) \right] \alpha_m \quad (51e)$$

$$\lambda'_r(\tau) = {}^n\mathcal{T}(\tau)\alpha_{\lambda_r}, \quad \lambda_r(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - v(-1) \right] \alpha_{\lambda_r} \quad (51f)$$

$$\lambda'_u(\tau) = {}^n\mathcal{T}(\tau)\alpha_{\lambda_u}, \quad \lambda_u(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - v(-1) \right] \alpha_{\lambda_u} \quad (51g)$$

$$\lambda'_v(\tau) = {}^n\mathcal{T}(\tau)\alpha_{\lambda_v}, \quad \lambda_v(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - v(-1) \right] \alpha_{\lambda_v} \quad (51h)$$

$$\lambda'_m(\tau) = {}^n\mathcal{T}(\tau)\alpha_{\lambda_m}, \quad \lambda_m(\tau) = \left[{}^n\mathcal{T}(\tau)\tilde{\mathcal{F}} - v(-1) \right] \alpha_{\lambda_m} \quad (51i)$$

Thus, all of the unknown coefficients are concatenated into the free variable vector

$$\phi = \begin{bmatrix} \alpha_r \\ \alpha_\theta \\ \alpha_u \\ \alpha_v \\ \alpha_m \\ \alpha_{\lambda_r} \\ \alpha_{\lambda_u} \\ \alpha_{\lambda_v} \\ \alpha_{\lambda_m} \\ \nu_r \\ \nu_\theta \\ \nu_u \\ \nu_v \\ f_f \end{bmatrix} \in \mathbb{R}^{9(n+1)+5 \times 1} \quad (52)$$

The initial and final orbit elements are specified as

$$e(t_0) = \begin{bmatrix} a_0 & e_0 & \omega_0 & f_0 \end{bmatrix} \quad (53)$$

$$e(t_f) = \begin{bmatrix} a_f & e_f & \omega_f \end{bmatrix} \quad (54)$$

where a is semi-major axis, e is eccentricity, ω is argument of periapsis, and f is true anomaly. The true anomaly of the targeted orbit is left free, and inclination and longitude of the ascending node are not needed for this coplanar case. Finally, the transfer time t_f , specific impulse I_{sp} , and thrust T are specified.

To obtain solutions, *Matlab*'s nonlinear solver *fsolve* is used to iteratively solve for the free variable vector ϕ , and an order $n = 250$ Chebyshev polynomial approximation is used for each state and costate. In all cases, the initial guess for all of the coefficients was 0, the costates were seeded at values of 0.3 (to avoid singularities at 0), and the final true anomaly was initiated at 2π . The problem is non-dimensionalized using time, mass, and distance units of 1 AU, 1000 kg, and 1 year, respectively. The initial orbit is always selected to be an Earth orbit, with varying initial true anomaly, and the final time for all problems is $t_f = 500$ days with a thrust of 0.6 mN. A continuation method is used to decrease ρ from 1 to $1e - 05$ in 8 steps.

An example trajectory is shown in Fig. 1, where the parameters are

$$\mathbf{e}(t_0) = \begin{bmatrix} 1 & 0.016 & 0 & 0 \end{bmatrix} \quad (55)$$

$$\mathbf{e}(t_f) = \begin{bmatrix} 4 & 0.5 & 120^\circ \end{bmatrix} \quad (56)$$

$$I_{sp} = 4000 \text{ s} \quad (57)$$

The corresponding throttle history for changing values of ρ is shown in Fig. 2.

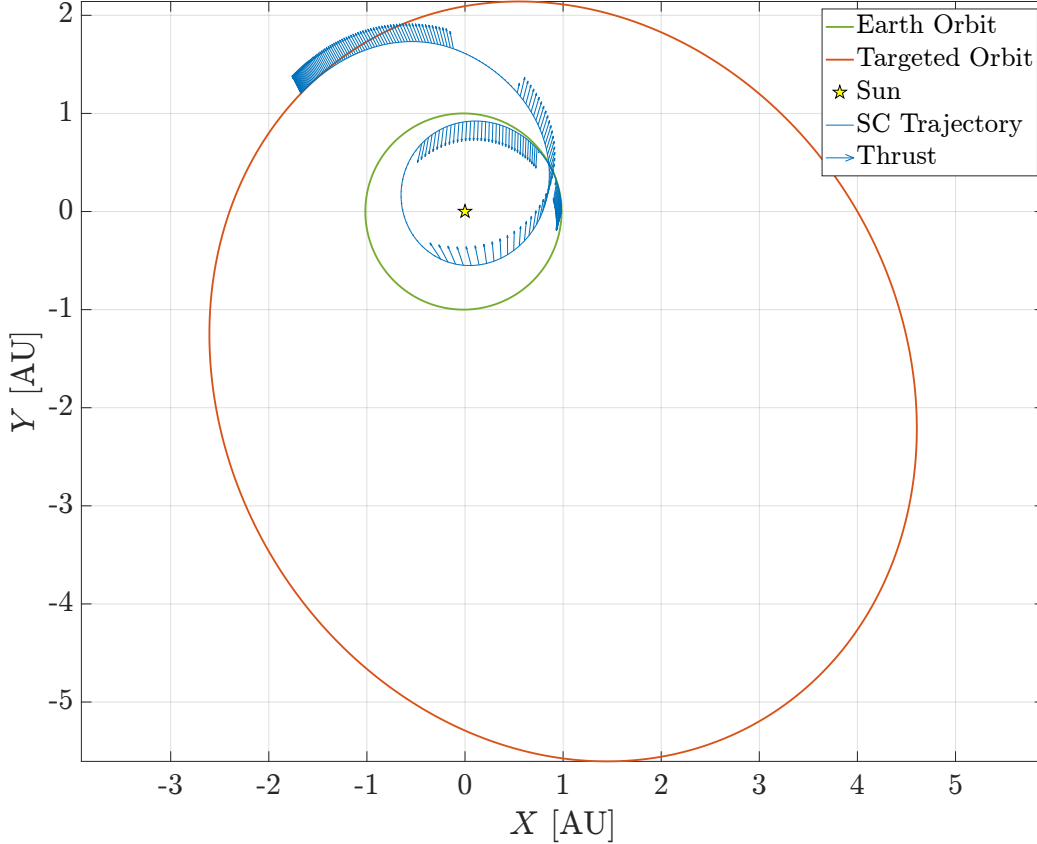


Figure 1: Optimal minimum fuel transfer orbit trajectory.

Once an initial solution has been found to the optimal control problem, neighboring solutions with similar parameters can be explored. First, consider the 1-dimensional extremal field map

$$f : \{I_{sp}\} \rightarrow \{\lambda_r \quad \lambda_\theta \quad \lambda_u \quad \lambda_v \quad \lambda_m\}^\top \quad (58)$$

To create this map, solutions to the optimal control problem presented will be found with varying values of $I_{sp} \in [2000, 5000]$ s and all other parameters held constant. To accommodate the orthogonal approximation method to be used, cosine spacing will be used with $M = 10$ nodes. At each node, the optimal control is found and the initial costates are recorded and used to create a 1-dimensional smooth orthogonal approximation of order $N = 10$. Fig. 3 shows both the nodes and approximation of each of the initial costates. A visual inspection confirms that the relationship between I_{sp} and the costates is relatively smooth, which provides confidence that the map can be

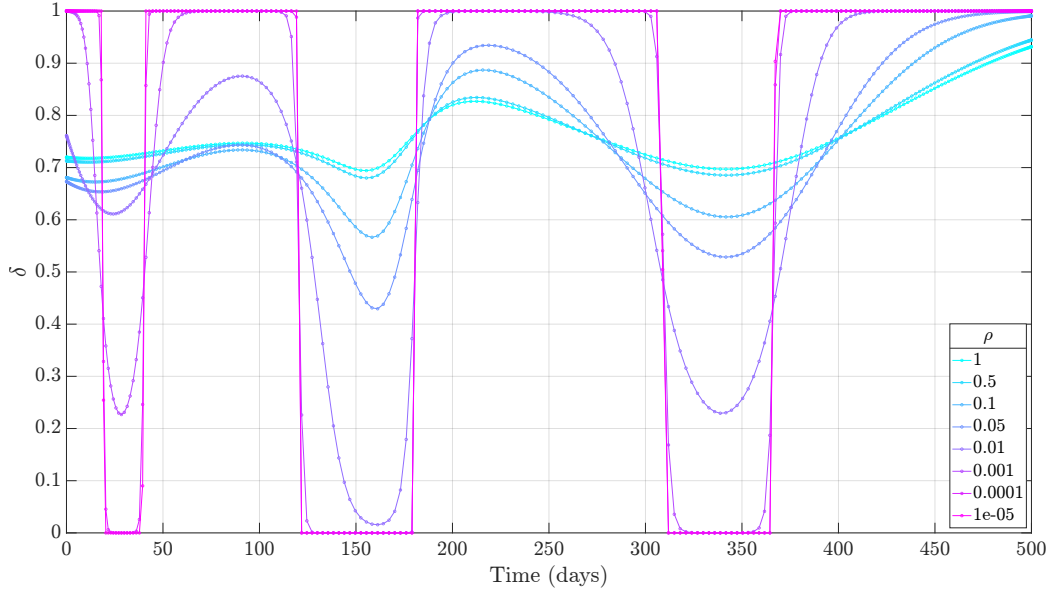


Figure 2: Throttle history with changing smooth parameter ρ .

approximated with a relatively low order. This observation will prove useful when creating maps of multiple variables. As the dimension of the parameter space increases, it is increasingly important to minimize the number of necessary sample points for each parameter because the total number of grid points increases exponentially. Considering that nonlinear solvers require significant computation time per solution, a lower number of grid points significantly decreases computation time.

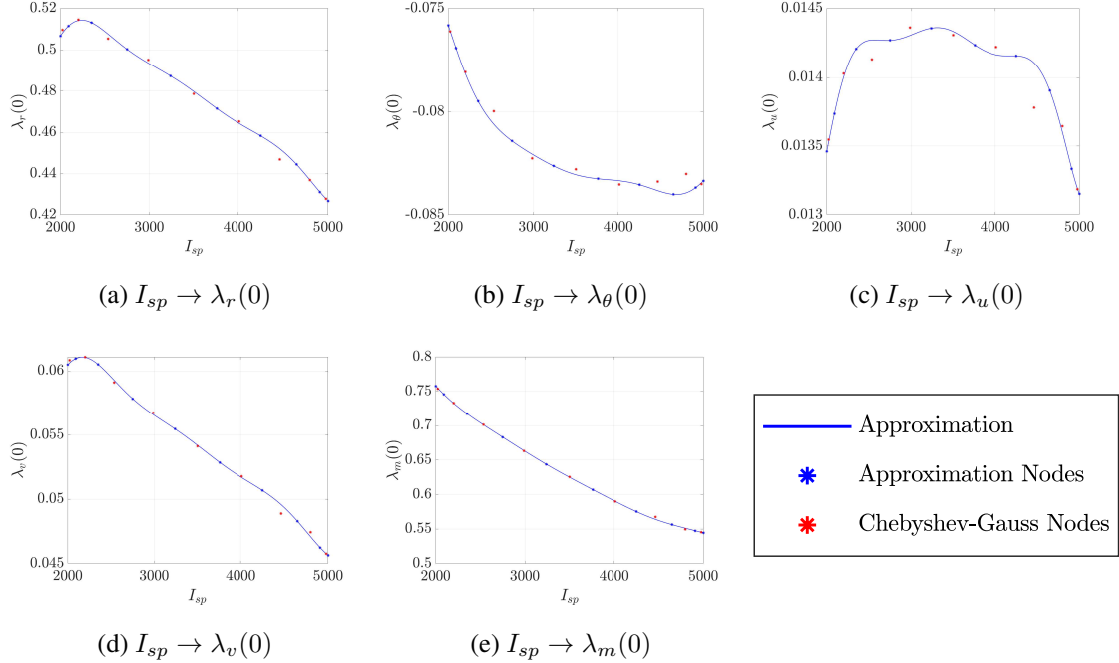


Figure 3: Extremal field map for $I_{sp} \in [2000, 5000]$ s

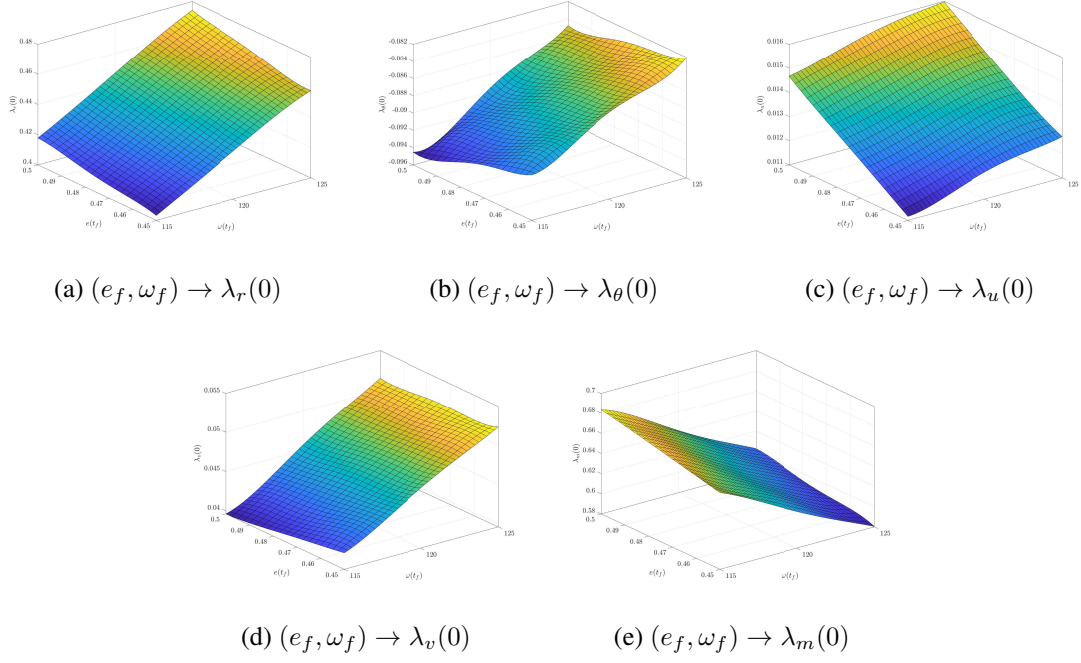


Figure 4: 2-dimensional extremal field map for $[e_f, \omega_f] \in [(0.45, 0.5), (115^\circ, 125^\circ)]$

Next, consider the extremal field map

$$f : \begin{Bmatrix} e_f & \omega_f \end{Bmatrix}^\top \rightarrow \begin{Bmatrix} \lambda_r & \lambda_\theta & \lambda_u & \lambda_v & \lambda_m \end{Bmatrix}^\top \quad (59)$$

which relates the final eccentricity and final argument of periapsis to the initial costates. Parameter values of $e_f \in [0.45, 0.5]$ and $\omega_f \in [115^\circ, 125^\circ]$ are considered. The ranges for these parameters must be constrained to those which correspond to reachable orbits, and therefore must be chosen more carefully than a parameter such as I_{sp} . Optimal control solutions are computed at points on a 2-dimensional cosine-spaced grid where the number of points in each dimension is restricted to $M_{e_f} = M_{\omega_f} = 5$. Fig. 4 shows smooth 2-dimensional approximations for the extremal field map for each costate. Again, the relationship over this interval is relatively smooth, and therefore a relatively low order ($N_x = N_y = 5$) approximation is sufficient.

To evaluate the validity of these maps, the approximations are compared to computed solutions. This comparison is made at points between CGL nodes, as the approximation should be accurate to machine precision at these points. To ensure this, the same intervals are sampled at the Chebyshev-Gauss (CG) nodes, defined in Eq. (11). These solutions are shown in red next to the interpolated solution in Fig. 3. These nodes do not overlap with the CGL nodes, and place grid points between each of the CGL nodes. For each costate, the L2 norm of the error at each CG node for approximations of the extremal field map

$$f : I_{\text{sp}} \rightarrow \lambda \quad (60)$$

is shown in Fig. 5. From the computed errors, it is apparent that higher-order approximations, at least over this interval, do not meaningfully impact the accuracy of the approximation; however,

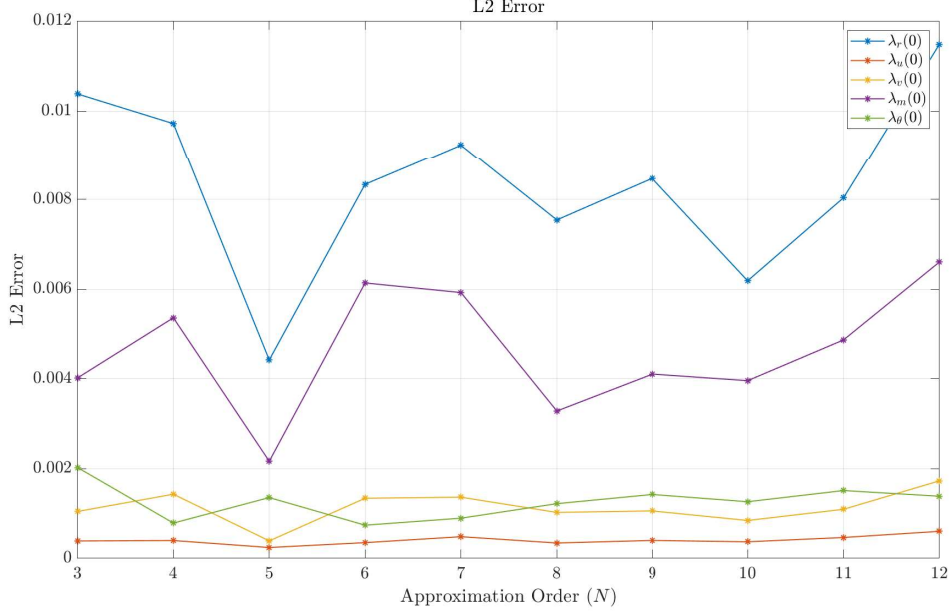


Figure 5: L2 error of extremal field map approximations at Chebyshev-Gauss node points.

with this level of accuracy, using the approximation value as an initial guess to find the true value will result in quick, low iteration convergence to the true value. Additionally, comparing the relative accuracy of each costate approximation, since the λ_r and λ_m have the highest error, it is likely that they are more sensitive or nonlinear than the other costates.

CONCLUSION

An attractive process is demonstrated for construction of extremal field maps. These extremal field maps provide a guide to the solution space in indirect optimal control problems. Once created, they reveal the behavior of the problem with respect to changing parameters, which provides valuable insight for mission design studies. For example, these could lead one to select one or more feasible low-thrust engines from a cluster of available options. Additionally, the costate data generated for large ranges of parameters is useful for improving the initial guess required for determining the solution to future OCPs with similar conditions. For future applications, this method could be used to describe parameter-solution relationships for common orbit transfers and trajectory optimization problems, leading to a large known solution space. Alternatively, the considered parameter space may motivate the creation of related field maps. Parameters could be swept along the course of the initial optimal trajectory, providing relevant data for station-keeping and trajectory correction. Ideally, the parameter space would be expanded dimensionally to include all possible static parameters and boundary conditions, thoroughly describing all possible perturbations of the initial problem.

The approximation methods demonstrated provided the necessary efficiency to repeatedly compute solutions to OCPs in a reasonable timeframe; this process is intensely parallelizable. ICC, while not the focus of this work, reduced computation time for each solution and reduced sensitiv-

ity of convergence with respect to the initial guess. In future work, generating an extremal field map in parallel to the repeated computation of OCP solutions, and using the map to inform the initial guess of the solution procedure as able, should provide additional improvements in efficiency.

This method, with continued optimization and exploration, shows promising potential to become a useful tool for indirect optimal control methods. The combination of proven approximation methods with favorable properties contributes heavily to the methods efficacy. While expansion on this work is certainly required, it demonstrates that approximated extremal field map methods are viable tools for understanding optimal control problems.

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